ShopFlow AWS Project

Application Load Balancer Design

Public Entry Point - Health Checks - Listener Rules

Module 9 of 18

Field	Value
Version	1.0
Date	June 2026
Region	ap-south-1 (Mumbai)
Author	Manish Kumar Sharma
Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

The Application Load Balancer is the public entry point for ShopVerse. It resides in both public subnets and accepts HTTP traffic from the internet on port 80. It forwards requests to EC2 or ECS tasks in private subnets on port 8080. Health checks poll /actuator/health every 30 seconds.

- Layer 7 (HTTP/HTTPS) load balancer -- routes by path, header, or host
- Spans two public subnets across two AZs -- ALB itself is highly available
- Health-checked targets: unhealthy targets removed from rotation automatically
- ShopVerse is stateless (JWT auth) -- sticky sessions not needed

2. ALB Configuration

Setting	Value	Reason
Type	Application Load Balancer	Layer 7 -- HTTP-aware routing
Scheme	internet-facing	Public entry point for users
Subnets	shopflow-pub-1a + shopflow-pub-1b	Spans 2 AZs for availability
Security Group	shopflow-alb-sg	Port 80/443 from internet
Idle Timeout	60 seconds	Close stale connections

3. Target Group

Setting	Value	Notes
Protocol	HTTP	ALB to EC2 is HTTP (TLS terminates at ALB)
Port	8080	Spring Boot server port
Health Check Path	/actuator/health	Spring Boot Actuator -- returns 200 when UP
Check Interval	30 seconds	Poll every 30 seconds
Healthy Threshold	2 successes	2 consecutive 200s -- Healthy
Unhealthy Threshold	3 failures	3 consecutive non-200s -- Unhealthy
Deregistration Delay	30 seconds	Finish in-flight requests before removing
Stickiness	Disabled	ShopVerse is stateless -- no sticky sessions

4. HTTP Listener

Listener	Protocol	Port	Default Action
HTTP Listener	HTTP	80	Forward to shopflow-tg-dev (Target Group)

For production: add an HTTPS listener on port 443 with an ACM certificate, and redirect HTTP 80 to HTTPS. ACM certificates are free.

5. Traffic Flow

Step	Component	Location	What Happens
1	User Browser	Internet	User navigates to shopflow.com
2	Internet Gateway	VPC Edge	Request enters AWS network
3	ALB	Public subnet	Receives HTTP:80, forwards to target
4	EC2 / ECS Task	Private subnet	Spring Boot processes on port 8080
5	RDS / Redis	Private subnet	App fetches data, reads cache
6	Response	Reverse path	Response flows back through ALB

6. CloudFormation Outputs

Export Name	Value
shopflow-alb-dns	ALB public DNS name (shopflow-alb-dev.ap-south-1.elb.amazonaws.com)
shopflow-alb-arn	ALB ARN (for WAF association in production)
shopflow-tg-arn	Target Group ARN (used by ECS service in Module 11)
shopflow-alb-http-listener-arn	HTTP Listener ARN (for additional routing rules)